

Lecture 2:
The Economic Approach (II)
ECON7720: Ecological and Environmental Economics

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Recap of Lecture 1

- The **scale and scope** of environmental problems (climate, air, water, biodiversity).
- The benchmark is the **efficient allocation**: maximising total surplus (CS + PS).
- **Demand** = MB; **supply** = MC; efficiency where $MB = MC$.
- **Static efficiency** and the **equimarginal principle**: equalise marginal net benefits across uses.
- **Dynamic efficiency**: maximise PV of net benefits; discounting and the **marginal user cost** of scarce resources.

Lecture Outline

1. Externalities and the Pigouvian tax/subsidy
2. Property rights, markets and the Coase theorem
3. Common-pool resources and the tragedy of the commons
4. Public goods and free riding
5. Other sources of inefficiency: monopoly, asymmetric information, government failure
6. The pursuit of efficiency: the courts and liability law

Externalities & Market Failure

What is an externality?

An **externality** arises when the welfare of an agent (a firm or a household) depends directly on the action of *another* agent, and that effect is **not transmitted through prices**.

- The decision-maker bears *private* costs and benefits, but ignores the costs or benefits imposed on *third parties*.
- The result: the **private** calculus and the **social** calculus diverge.

The “missing market”

For the spillover (smoke, noise, a cleaner neighbour, an idea) there is **no market and no price**. With no price, the agent receives no signal to account for it $\Rightarrow$ the competitive equilibrium is **inefficient**. This is the canonical form of **market failure**.

Recall the **Great Smog** from **Lecture 1**: cheap coal smoke imposed thousands of deaths on Londoners that no price reflected, and caused the 1956 Clean Air Act.

A taxonomy of externalities

Sign:

- **Negative** externality → harm imposed (pollution, congestion). Too *much* of the activity.
- **Positive** externality → benefit conferred (vaccination, R&D). Too *little*.

Source:

- **Production** externality → a firm's output affects others (e.g. factory effluent).
- **Consumption** externality → a household's consumption affects others (e.g. smoking, driving).

Private vs. social

Costs

$$\underbrace{\text{MSC}}_{\text{marginal social cost}} = \underbrace{\text{MPC}}_{\text{marginal private cost}} + \underbrace{\text{MEC}}_{\text{marginal external cost}}$$

Benefits

$$\underbrace{\text{MSB}}_{\text{marginal social benefit}} = \underbrace{\text{MPB}}_{\text{marginal private benefit}} + \underbrace{\text{MEB}}_{\text{marginal external benefit}}$$

- The market clears where $\text{MPB} = \text{MPC}$ (private optimum).
- Efficiency requires $\text{MSB} = \text{MSC}$ (social optimum).
- For a **negative production** externality, $\text{MEC} > 0$ so $\text{MSC} > \text{MPC}$: the supply curve the firm uses lies *below* the true social cost curve.

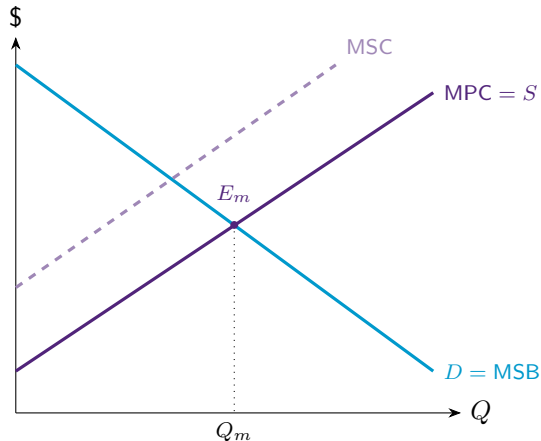
Negative production externality (1/2)

Setup

A competitive market. Each unit of output also imposes external cost MEC (e.g. effluent harming a downstream fishery).

- $S = MPC$: the firm's private supply.
- $MSC = MPC + MEC$ sits above S .
- $D = MPB = MSB$ (no consumption spillover here).

The market settles at E_m where D meets MPC .

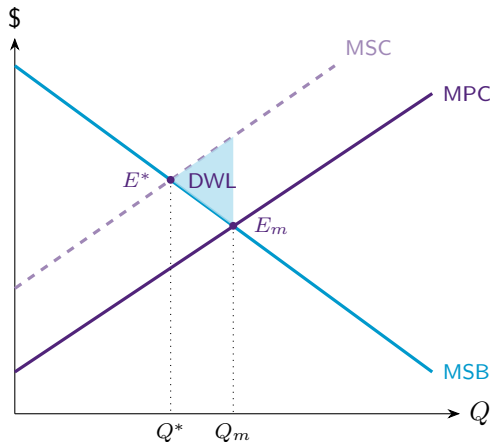


Negative production externality (2/2) — over-production & DWL

The inefficiency

At Q_m the *last* units have $MSC > MSB$: society pays more than the unit is worth. The efficient point is E^* where $MSB = MSC$.

- Market over-produces: $Q_m > Q^*$.
- The **deadweight loss** is the shaded triangle between MSC and MSB over $[Q^*, Q_m]$.



The Pigouvian tax (Pigou, 1920)

Levy a per-unit tax t that makes the firm **pay for the externality**. The firm's effective marginal cost becomes $\text{MPC} + t$. To restore efficiency we need that augmented cost to coincide with MSC *at the optimum*:

$$t^* = \text{MEC}(Q^*)$$

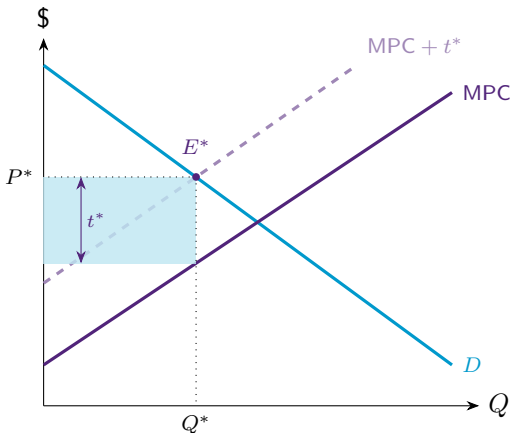
the marginal external cost **evaluated at the social optimum** Q^* .

- With the tax, private supply shifts up to $\text{MPC} + t^*$, which passes through E^* . Firms now choose Q^* *voluntarily*: the externality is **internalised**.
- The wedge between price and private cost is exactly the unpriced damage.
- **Tax revenue** $= t^* \times Q^*$ accrues to the government.

The Pigouvian tax — geometry

The tax lifts MPC to $MPC + t^*$, which crosses D at E^* .

- Output falls $Q_m \rightarrow Q^*$.
- Shaded rectangle = tax revenue $t^* \times Q^*$.
- The original DWL triangle is **eliminated**.



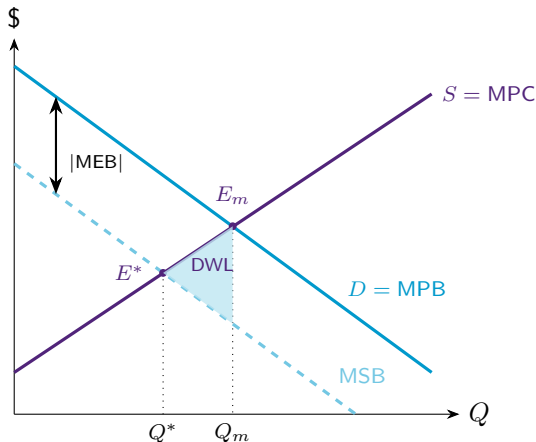
Negative consumption externality (1/2)

Example — Driving & congestion

Each driver slows *everyone else*: a **negative** external benefit ($MEB < 0$).

$$MSB = MPB + \underbrace{MEB}_{< 0}$$

- Externality on the **demand** side: MPB lies *above* MSB.
- Market at E_m ($MPB = MPC$); optimum at E^* ($MSB = MPC$).
- $Q_m > Q^*$: **over-consumption**.



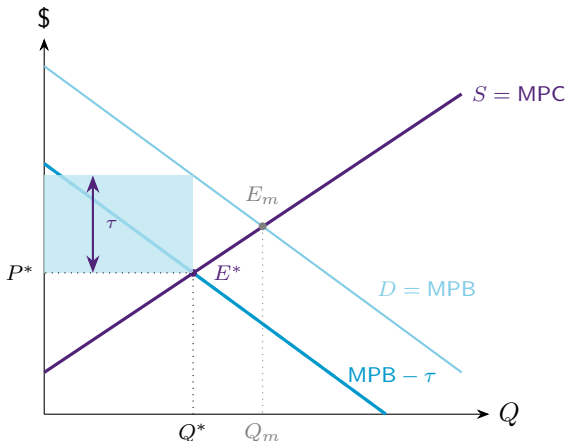
Negative consumption externality (2/2)

Example — Driving & congestion

A congestion charge $\tau = |\text{MEB}(Q^*)|$ makes each driver face the full social cost of her trip.

- The tax shifts the private demand curve down from MPB to $\text{MPB} - \tau$, which coincides with MSB at Q^* .
- Drivers now choose Q^* voluntarily: the externality is internalised.
- The original DWL triangle is eliminated.
- Tax revenue $= \tau \times Q^*$ can fund public transport.

Same Pigouvian logic as the production externality (slides 10–11), but applied to the demand side.

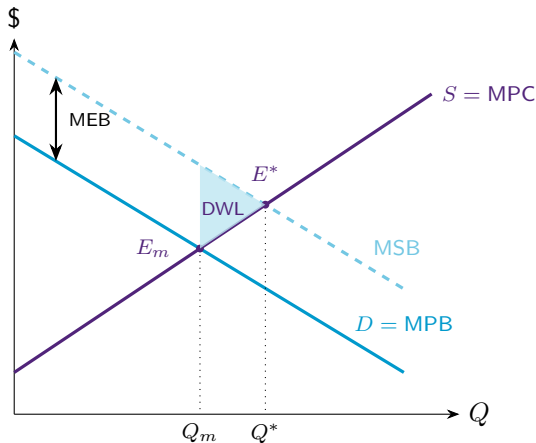


Positive externality — under-production (1/2)

Examples

Vaccination (herd immunity), R&D (knowledge spillovers), on-farm conservation.

- Here $MSB = MPB + MEB$ with $MEB > 0$.
- Externality on the **demand** side: MSB lies *above* MPB.
- Market at E_m ($MPB = MPC$); optimum at E^* ($MSB = MPC$).
- $Q_m < Q^*$: **under-production**.



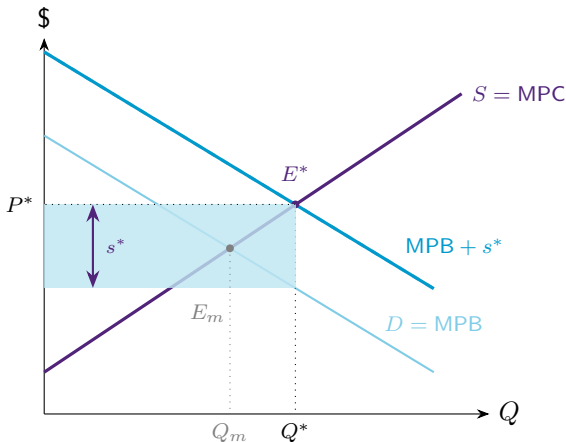
Positive externality — the Pigouvian subsidy (2/2)

The corrective subsidy

A subsidy $s^* = \text{MEB}(Q^*)$ makes each agent face the full social benefit of her activity.

- The subsidy shifts the private demand curve up from MPB to $\text{MPB} + s^*$, which coincides with MSB at Q^* .
- Agents now choose Q^* voluntarily: the externality is internalised.
- The original DWL triangle is eliminated.
- Subsidy cost = $s^* \times Q^*$ borne by the government.

Symmetric to the tax: tax the bad, subsidise the good, each set to the external effect at the optimum.



Five consequences of unpriced negative externalities

When external costs go unpriced, the market delivers. . .

1. Too much output: the commodity is over-produced ($Q_m > Q^*$).
2. Too much pollution: residuals exceed the efficient level.
3. Prices that are too low: the product price does not reflect the full social cost, so consumers over-consume.
4. No incentive to innovate: because pollution is free, firms have no reason to develop cleaner production processes or to reduce emissions per unit of output.
5. Recycling and reuse discouraged: disposing of residuals costs nothing, so recovering and reusing materials is uncompetitive even when it would be socially worthwhile.

Property Rights, Markets & the Coase Theorem

Competitive equilibrium maximises net benefits (recall from Lecture 1)

The central efficiency result

In a competitive market, the equilibrium price and quantity (P^*, Q^*) **maximise the net benefits to society**: the sum of consumer surplus and producer surplus.

- The **demand** curve traces marginal (private) benefit; the **supply** curve traces marginal (private) cost.
- At Q^* marginal benefit = marginal cost: the last unit traded is exactly worth its cost.
- Producing less leaves valued units unrealised; producing more adds units that cost more than they are worth.

First Welfare Theorem (intuition)

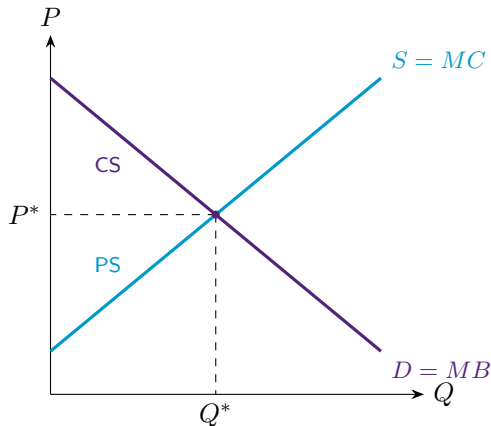
Any competitive equilibrium is Pareto efficient: no reallocation can make someone better off without making another worse off.

Surplus at the competitive equilibrium (recall from Lecture 1)

Consumer surplus (CS): area below D , above P^* , value to buyers net of what they pay.

Producer surplus (PS): area above S , below P^* , revenue net of variable cost.

Net benefit = $CS + PS$ is largest at Q^* . Any other quantity shrinks the combined triangle (a deadweight loss).



When do markets deliver efficiency?

The efficiency conditions (all must hold)

- Price-taking competition: many buyers and sellers; no agent moves the price.
- Well-defined property rights: every resource is owned, and ownership is clear.
- No externalities or public goods: all costs and benefits fall on the transacting parties.
- Good information: agents know prices, qualities and consequences.

Why this matters for the environment

Environmental problems are precisely **where these conditions fail**: clean air and fisheries are unpriced and unowned; pollution is an externality; ecosystem information is incomplete. The market outcome is then **not** efficient, the rationale for policy.

Property rights

Definition

A **property right** is not the physical thing but a **bundle of entitlements** defining the owner's rights, privileges and limitations in using a resource.

- Entitlements may be split: the right to *use*, to *exclude*, to *sell*, to receive *income*, to *pass on by inheritance*.
- Example: a farmer may own the right to grow crops on land but **not** the right to the minerals beneath, or to divert an unlimited share of river water.

Why the structure matters

The **structure** of these rights shapes incentives, and therefore how efficiently a resource is used and conserved.

Three features of an efficient property-rights structure

Exclusivity

All benefits and costs accrue **to the owner**. Forces the owner to weigh *all* consequences; no costs shifted onto others (no externality).

Example: a private forest owner reaps the gain from replanting trees they will later harvest.

Transferability

Rights can be **voluntarily traded**. Lets resources flow to their highest-valued use through trade.

Example: a water entitlement can be sold to a higher-value user, moving the resource to its best use.

Enforceability

Rights are **secure** and protected against unauthorised use or appropriation. Gives owners confidence to invest, conserve and trade. Without it, rights are hollow.

Example: a registered land title that police and courts will defend against trespass.

Where all three hold, self-interested owners are led to use resources efficiently. Most environmental degradation traces to a **failure of one or more** of these features.

Property-rights regimes

Private property

Rights held by an individual or firm.

Example: a privately managed plantation forest, harvested sustainably.

State property

Rights held by government on behalf of citizens.

Example: a national park managed for public conservation.

Common property

Rights held **jointly by a defined group** that can exclude outsiders and set internal rules.

Example: a village-managed grazing common or community fishery.

Open-access

No one owns or can exclude; the resource is open to all.

Example: an unregulated open-ocean fishery.

The Coase Theorem (Coase, 1960)

If property rights are **well defined and tradeable**, and **transaction costs are low**, then private bargaining between affected parties will reach the **efficient allocation**, **regardless of who initially holds the right**.

- The **assignment** of the right changes only the **distribution** of wealth (who pays whom).
- It does **not** change the efficient quantity of pollution / resource use.
- Profound implication: under these conditions, government need only **define and protect rights**; the parties sort out the rest.

When Coasean bargaining breaks down

Transaction costs are the binding constraint

Coase assumed them **low**. In environmental settings they are often **prohibitive**.

- **Many parties**: millions affected by air pollution or climate change; bargaining is infeasible.
- **Holdouts**: when many must agree, a single party can block the deal to extract more.
- **Asymmetric information**: parties cannot verify true damages or true abatement costs.
- **Free riding**: with public goods, each party hopes others pay; the deal collapses.

Implication

For large-scale externalities, decentralised bargaining fails $\Rightarrow$ a role for **government policy** (taxes, standards, markets).

Policy implication

Clarify and assign rights, then let markets work

The Coasean insight survives even when full bargaining fails: government can **create well-defined, tradeable rights** so that a smaller, structured market does the allocating.

- **Emissions trading**: define a fixed number of tradeable allowances (a cap). Firms with low abatement costs sell; high-cost firms buy. Pollution is cut **where it is cheapest**.
- **Water markets**: define volumetric entitlements that can be traded, so water flows to its highest-value use in dry years.

Common logic

Both *manufacture* exclusivity and transferability where nature provided none, turning an open-access problem into a priced, ownable one.

Common-Pool Resources & the Tragedy of the Commons

Two attributes that classify all goods

- **Excludable**: can a non-payer be feasibly kept from using it? (a fence, a password, a fishing licence)
- **Rival** (subtractable): does one person's use diminish what is left for others?

	Rival	Non-rival
Excludable	Private goods food, a car, a privately caught fish	Club / toll goods cable TV, a toll road, a private park
Non-excludable	Common-pool resources ocean fishery, aquifer, open rangeland	Public goods national defence, a stable climate

Common-pool resources

Non-excludable but rival

- **Non-excludable**: hard or costly to keep additional users out; the ocean has no fences.
- **Rival**: every unit harvested is a unit gone; the fish I land is not available to you.

Examples

- Marine fisheries
- Groundwater aquifers
- Forests and timber on unmanaged land
- The atmosphere as a sink for emissions
- Open rangelands and common pastures

The combination “rival + non-excludable” is exactly the recipe for the tragedy of the commons.

The tragedy of the commons (Hardin, 1968)

The key question

If a resource belongs to **nobody** (or equivalently, to **everybody**), who has the incentive to conserve it?

Imagine n herders sharing a pasture. Each one who adds an animal:

- **Private benefit**: the herder captures the **full** value of the animal (+1).
- **Private cost**: the herder bears only $1/n$ of the damage to the pasture.
- **Social cost**: the *total* damage to the pasture (-1) is spread across all n herders, but each herder ignores the $(n-1)/n$ that falls on others.

The herder compares **private benefit** to **private cost**: $1 > 1/n \Rightarrow$ add the animal.

But the **social cost** (the full damage) exceeds the private benefit at the margin. Since every herder faces the same calculus, **all of them** add animals $\Rightarrow$ **over-exploitation**.

Why is it a “tragedy”?

Over-exploitation, not necessarily destruction

The tragedy is **not** that the pasture is necessarily destroyed. It is that the **rent**, the value the pasture could generate above costs, is progressively **wasted**, because each herder's private cost understates the true social cost.

- More herders $\Rightarrow$ more rent dissipated.
- As $n \rightarrow \infty$: rent $\rightarrow 0$, full **open access**.

Why doesn't a sole owner over-exploit?

A sole owner ($n = 1$) bears the *full* social cost of each additional animal. Her private cost *equals* the social cost, so she maximises **rent** by stopping where marginal benefit = marginal social cost, the same solution as the social planner. The tragedy arises precisely because shared access **drives a wedge** between private and social cost.

In extreme cases (many users, slow regeneration, no escape), over-exploitation can push the resource past an irreversible threshold and cause **collapse**.

Planner vs. open access

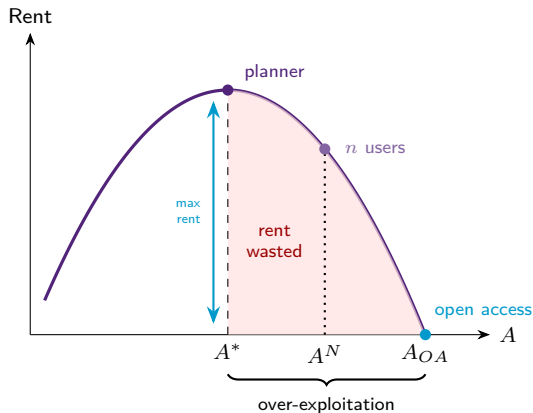
Social planner / sole owner

- Bears the **full social cost** of each animal.
- Stops where $MB = MSC$.
- Chooses $A^* \Rightarrow$ **rent is maximised**.

Open access ($n \rightarrow \infty$)

- Each user ignores **all** social cost.
- Entry continues while average benefit $>$ cost.
- Pushes to $A_{OA} \Rightarrow$ **rent = zero**.

Any finite n gives a Nash equilibrium **between** A^* and A_{OA} . The larger n , the more rent is destroyed.



An open-access fishery — setting up the model

Effort, harvest and value

The same logic as the herder model, now applied to a fishery: **effort** E replaces animals A , and the hump-shaped harvest replaces $v(A)$.

Let **effort** E = number of boats (or boat-days). The biology gives a harvest that first rises, then falls as the stock is depleted, so the **total value of harvest** $TB(E)$ is hump-shaped:

$$TB(E) = (\text{price}) \times \text{harvest}(E), \quad TB'(E) > 0 \text{ then } < 0.$$

Each unit of effort costs the same c , so **total cost** is linear:

$$TC(E) = cE.$$

Resource rent = profit = $TB(E) - TC(E)$. The next two slides find (i) the *efficient* effort that maximises rent and (ii) the *open-access* effort that dissipates it.

Efficient effort vs. open access

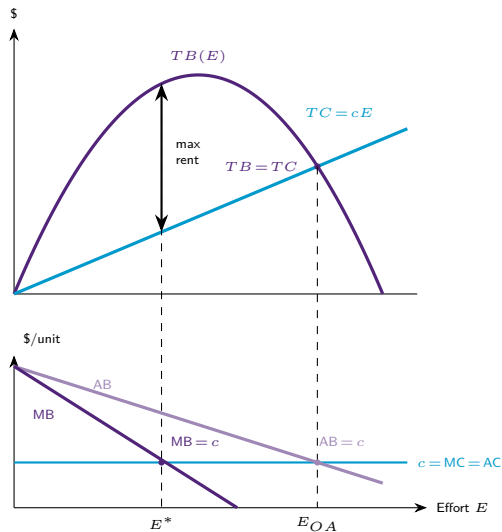
Efficient (E^*)

- Planner sets $MB = c$.
- Rent ($TB - TC$) is maximised.

Open access (E_{OA})

- Entrants compare average benefit to cost: $AB = c$.
- Entry continues until $TB = TC$: rent = 0.
- $E_{OA} > E^*$: too many boats.

$MB < AB$ over the relevant range: each entrant ignores the stock externality she imposes on others.



Why open access over-exploits — the externality

Average vs marginal

A new entrant compares her catch to *her* cost; she looks at the **average** product of effort, which is still positive at E^* .

$$\underbrace{\text{efficient rule}}_{MB=c} \quad \text{vs} \quad \underbrace{\text{open-access rule}}_{AB=c}$$

Because $MB < AB$ over the relevant range, the open-access entry rule pushes effort **past** E^* until rent hits zero.

The missing cost

Each extra boat crowds the stock and lowers *everyone's* catch, a **stock/congestion externality**. The entrant ignores this external cost, so the private cost of effort understates the social cost. **Excessive effort** is the result.

Formalising the tragedy — the n -herder game (1/2)

Setup

n identical herders share a pasture. Herder i chooses $a_i \geq 0$ animals. Total animals: $A = \sum_{j=1}^n a_j$. Value per animal **falls with congestion**: $v(A)$, with $v' < 0$. Cost per animal: c (constant).

Rent $= A[v(A) - c]$: the total value the pasture generates above costs. The question is: *who maximises it?*

Individual herder (Nash)

$$\max_{a_i} \pi_i = a_i [v(A) - c]$$

$$\text{FOC: } v(A) + a_i v'(A) = c$$

Social planner (efficient)

$$\max_A W = A [v(A) - c]$$

$$\text{FOC: } v(A) + A v'(A) = c$$

Formalising the tragedy — the n -herder game (2/2)

The herder vs the planner

The herder accounts for only **her own** a_i times v' . The planner accounts for A times v' — the **full** social marginal cost. In a **symmetric Nash equilibrium** $a_i = A/n$:

$$\underbrace{v + A v' = c}_{\text{planner}} \quad \text{vs} \quad \underbrace{v + \frac{A}{n} v' = c}_{\text{Nash}}$$

Since $v' < 0$, the Nash condition subtracts **less** $\Rightarrow$ satisfied at a **larger** A .

- $n = 1$: herder = planner. **No tragedy.**
- n finite: $A^* < A^N < A_{OA}$. Rent partly wasted.
- $n \rightarrow \infty$: $\frac{1}{n} \rightarrow 0$, Nash $\rightarrow v(A) = c$. **Full open access**, rent = 0.

The **wedge** between private and social marginal cost is $(1 - \frac{1}{n}) A |v'|$, growing in n .

The dynamic dimension — stock and collapse

It is not just a static over-supply of effort

The static model assumes the stock has settled. But effort feeds back on the **stock** X :

$$\dot{X} = \underbrace{g(X)}_{\text{natural growth}} - \underbrace{h(E, X)}_{\text{harvest}}.$$

- If harvest persistently exceeds growth, the stock **shrinks**.
- Many populations have a **minimum viable stock**: below it, growth $g(X)$ falls faster than harvest, and the resource **collapses**, possibly to extinction.
- Open access provides no brake: by the time rent is zero the stock may already be critically low.

This is the **stock externality** in motion: today's effort lowers *tomorrow's* productivity for everyone.

Historical case — the American bison

- Pre-1800: an estimated 30–60 million bison roamed the Great Plains.
- Open access + a hide market + rail access $\Rightarrow$ industrial-scale slaughter in the 1870s.
- By the mid-1880s: fewer than 1,000 animals survived.

No one owned the herd; each hunter who held back simply left the animal for the next. **Open-access dynamics** (Lueck, 2002).

Recovery through exclusion

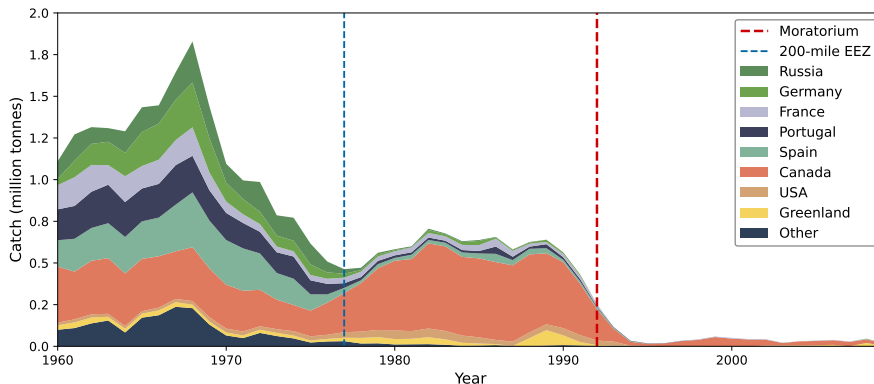
1889: only 1,091 left. Federal refuges (1907–13) + private ranching re-created **exclusion**. Today: $\sim$ 350,000 bison, mostly in private herds.



Pile of bison skulls waiting to be ground for fertiliser, c. 1870s.

Source: Burton Historical Collection, Detroit Public Library (public domain).

Historical case — the Atlantic cod



Data: NAFO (Northwest Atlantic Fisheries Organization) STATLANT 21B.

- For centuries, catches below 300,000 t/yr (sustainable). Post-WWII industrial fishing: peak 1.8 Mt.
- Spawning biomass crashed 93%: 1.6 Mt (1962) → 100,000 t (1992).
- 30+ years later: stock at ~12% of 1960s level. Moratorium lifted 2024.

Elinor Ostrom, *Governing the Commons* (Nobel 2009)

The commons need not be a tragedy

The prevailing view after Hardin (1968) emphasised two solutions: **privatisation** or **government regulation**. Ostrom demonstrated an **alternative way**: communities can govern CPRs *themselves*. She documented hundreds of **long-lived** cases (Swiss alpine pastures, Spanish irrigation *huertas*, Japanese forests, coastal fisheries) that avoided collapse without either.

Why does it work?

- Users have **local knowledge** outsiders lack; **repeated interaction** sustains cooperation through reputation and reciprocity.
- **Monitoring** is cheap when users observe each other daily; **graduated sanctions** deter without destroying trust.
- Success follows a set of **design principles** (next slide).

This institutions-first approach earned Ostrom the 2009 Nobel Memorial Prize in Economic Sciences — the first woman to receive it.

Ostrom's design principles

1. **Clearly defined boundaries:** who may use the resource, and what its physical limits are.
2. **Congruence:** rules for appropriation and provision fit local conditions, costs, and benefits.
3. **Collective-choice arrangements:** most users can participate in modifying the rules that bind them.
4. **Monitoring:** monitors who audit conditions and behaviour are accountable to the users.
5. **Graduated sanctions:** violators face penalties that escalate with the seriousness and context of the offence.
6. **Conflict-resolution mechanisms:** cheap, rapid arenas for resolving disputes among users .
7. **Minimal recognition of rights to organise:** external authorities do not undermine the community's right to make its own rules.
8. **Nested enterprises:** for large systems, governance is layered in multiple nested tiers.

Source: Ostrom (1990, *Governing the Commons*, pp. 90–102).

Public Goods

What is a public good?

Two defining properties

A pure public good is simultaneously:

- **Non-excludable**: once provided, no one can be (cheaply) prevented from consuming it.
- **Non-rival** (indivisible): one person's consumption does *not* diminish the amount available to others. Marginal cost of an extra user is zero.

Why markets struggle

Non-excludability undermines the ability to charge a price; non-rivalry means the *efficient* price for an extra user is zero. Both pull against a functioning private market.

Environmental public goods

Classic examples

National defence, lighthouses, basic research, a flood-control levee.

Environmental examples — the core of this course

- **Clean air** and air-quality regulation.
- **Climate stability**: a stable atmospheric carbon stock.
- **Biodiversity** and habitat / species preservation.
- **Ecosystem services**: pollination, water purification, carbon sequestration, storm protection.

Many of these are *global* public goods: their benefits cross borders, which sharpens the free-riding problem (returned to later).

Biological diversity

What is biodiversity?

- **Genetic diversity**: variation among individuals *within* a species (resilience to disease, adaptation to change).
- **Species diversity**: the number and variety of species in a community.
- **Ecosystem diversity**: the range of distinct ecosystems (wetlands, coral reefs, forests) across a landscape.

Why it is a public good

- Biodiversity provides **non-rival, non-excludable** services: ecological stability, pollination, pest control, potential sources of food, raw materials and medicines.
- Its value extends well beyond any single user's willingness to pay: **existence value** (people value knowing species survive) and **option value** (future uses we cannot yet foresee).
- Will private markets produce the efficient amount of biological diversity? **No**: the free-rider problem applies in full. Each landholder under-conserves, and species loss is largely **irreversible**.

Efficient provision

Everyone consumes the *same* quantity

Because a public good is non-rival, one unit provided is enjoyed by *all* consumers simultaneously. The relevant aggregate is therefore the **sum of marginal benefits at each quantity**.

- **Private good**: at a price, each person picks her own quantity $\Rightarrow$ **horizontal** summation of demands (add quantities at each price).
- **Public good**: quantity is common, willingness-to-pay differs $\Rightarrow$ **vertical** summation of marginal-WTP curves (add prices at each quantity).

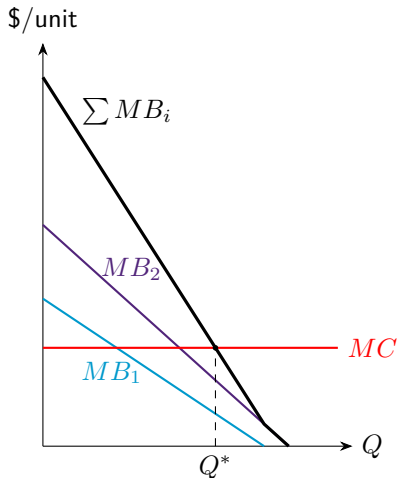
The Samuelson condition (Samuelson, 1954)

Efficient provision sets the sum of marginal benefits equal to marginal cost:

$$\sum_{i=1}^n MB_i(Q) = MC(Q).$$

Intuition: for a private good, one unit goes to *one* person, her MB alone must cover MC. For a public good, one unit serves *everyone at once*, so *all* their MBs together must cover MC.

Vertical summation of demand



Two consumers

Each curve gives marginal **willingness to pay** for the public good at quantity Q .

Add prices vertically

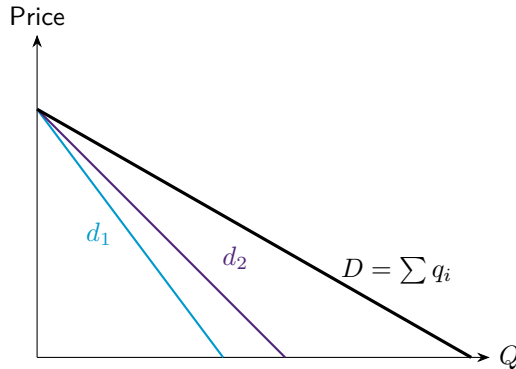
At each Q , **stack the heights**:

$\sum MB_i = MB_1 + MB_2$. The aggregate (black) curve lies **above** each individual curve.

Efficient quantity Q^* where
 $\sum_i MB_i = MC$.

The curve **kinks** where the lower-valuation consumer drops out ($MB_1 = 0$).

Contrast — a private good (horizontal summation)



Add quantities horizontally

At each **price**, sum the quantities each consumer demands. The market demand lies to the **right** of each individual curve.

Why? Each person freely chooses *her own* quantity at a common price. This is the opposite of a public good, where the **quantity** is common and the prices (valuations) differ.

Why the summation direction differs

The key distinction comes down to one question: **who chooses what?**

	Private good (e.g. apple)	Public good (e.g. park)
What's shared?	The price	The quantity
What differs?	Each person's quantity	Each person's valuation
How to aggregate?	Add quantities → horizontal	Add valuations → vertical
Why?	One apple goes to one person	One park serves everyone at once

One unit of a public good serves **everyone simultaneously**, so the question is not “how many units total?” but “how much is this one shared unit worth to **everyone combined**?”

Free riding and under-provision

The free-rider problem

Because the good is non-excludable, each person can enjoy it whether or not she pays. The privately rational choice is to let others pay and consume for free.

- Each agent equates her private marginal benefit to her marginal cost of contributing.
- But the social marginal benefit $= \sum_i MB_i$ greatly exceeds any one person's MB_i .
- Result: private contributions stop far short of Q^* : systematic under-provision.

Externality view

Each contributor confers an uncompensated positive externality on everyone else. Positive externalities $\Rightarrow$ too little of the activity in a free market.

The demand-revelation problem

The core difficulty

Efficient provision requires knowing each person's true MB_i . But if charges are tied to stated valuations, everyone has an incentive to **understate** (free-ride) $\Rightarrow$ true preferences are hidden.

Lindahl prices (Lindahl, 1919)

If we knew each person's MB_i , we could charge a **personalised price** $\tau_i = MB_i(Q^*)$ with $\sum_i \tau_i = MC$. Each person would then voluntarily demand Q^* , efficient provision achieved.

Why it fails: implementing Lindahl prices requires exactly the information that free-riding conceals. It is a theoretical benchmark, not a practical tool, but it clarifies *why* public-good provision is hard.

Provision mechanisms

Recall: free markets won't provide public goods efficiently (free-riding).

Government provision financed by taxation

The state supplies the good and compels payment via taxes, sidestepping non-excludability. Challenge: knowing the efficient Q^* without market signals.

Subsidies and incentives

Subsidise private actions that generate public benefits (e.g. tax credits for habitat restoration), internalising the positive externality.

Voluntary provision and clubs

Philanthropy, NGOs, and **clubs** that add excludability (membership) to fund an otherwise non-excludable good: partial, often under-provided, solutions.

Payments for ecosystem services

Pay landholders for the public benefits their land provides (watershed protection, carbon). A market-like instrument we return to later in the course.

Global public goods and international free riding

Climate stability and biodiversity are global public goods

A tonne of CO₂ abated benefits every country; a preserved species or rainforest benefits the whole world. Benefits are non-rival and non-excludable across **nations**.

- Each country bears the **full cost** of its own abatement but captures only a **small share** of the global benefit.
- The free-rider game now has *many* players and **no world government** to tax or compel.
- Hence the equilibrium tendency: **too little abatement**.

International treaties (e.g. the Paris Agreement) attempt to build the cooperation that markets cannot, but enforcement remains the binding constraint.

Other Sources of Market Failure

A wider map of market failure

The First Welfare Theorem says competitive markets are efficient **only under strong conditions**: price-taking behaviour, complete and symmetric information, well-defined property rights, no externalities or public goods.

When those conditions fail?

- **Imperfect competition**: firms with market power set price above marginal cost.
- **Asymmetric information**: one side knows more than the other.
- **Incomplete markets**: markets for some goods simply don't exist (e.g. a stable climate, biodiversity, future generations' preferences).
- **Transaction costs**: costs of negotiating, monitoring, and enforcing trades can prevent otherwise efficient bargains (recall Coase).

Each has a distinctive **environmental** dimension.

But beware: the corrective policy can itself misfire, **government failure**.

Imperfect competition

The decision rule

A competitive firm is a price-taker: it expands until $P = MC$. A monopolist faces the *whole* downward-sloping demand curve, so selling one more unit lowers the price on *all* units sold.

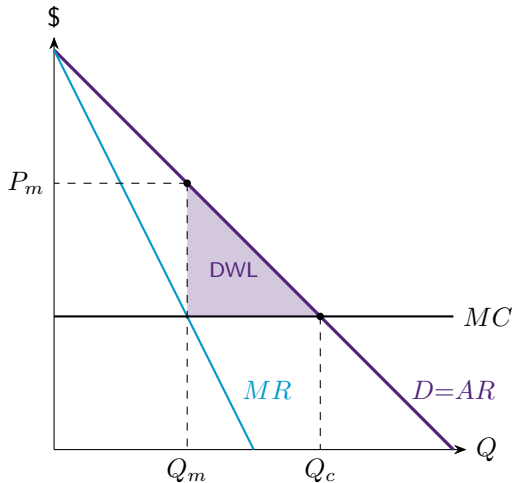
Profit-maximising choice

The monopolist sets $MR = MC$, which gives

$$Q_m < Q_c \quad \text{and} \quad P_m > P_c.$$

Output is *restricted* and price *raised* relative to the competitive benchmark. The lost mutually-beneficial trades are a *deadweight loss (DWL)*.

The deadweight loss of monopoly



- Monopoly output Q_m where $MR = MC$.
- Price P_m read **up to demand**, above MC .
- Competitive output Q_c where $D = MC$.

The welfare loss

The shaded triangle is value to consumers (D) that exceeds the cost of supply (MC) on units **not produced**. It is a pure **efficiency loss**, not a transfer.

Market power and the environment

The “conservationist monopolist”

For an **exhaustible resource** the story can reverse. A monopolist restricts output to keep prices high, which means **extracting more slowly** and leaving more in the ground for the future.

- Hotelling logic: monopoly tilts the extraction path **toward conservation**.

But the slowdown is driven by profit, not stewardship, and the static DWL remains. Do not read it as an argument *for* monopoly.

Cartels

Cartels approximate monopoly

A **cartel** coordinates output across producers to mimic a monopolist, with OPEC in oil being the classic case.

Two-sided environmental effects

- **Higher prices** curb consumption and slow depletion, and make substitutes (renewables, efficiency) more competitive.
- But cartels are **unstable**: members cheat, and price spikes can trigger a scramble for high-carbon alternatives (coal, sands).

Asymmetric information

The problem

Efficient trade assumes both sides know the relevant facts. When information is **asymmetric**, the better-informed party can exploit the gap, and good trades may fail to happen.

Two canonical forms, distinguished by *when* the hidden thing matters:

Hidden information (before the deal)

A characteristic is unobservable **ex ante**: the true quality of a product, the real contamination of a site. Leads to **adverse selection**.

Hidden action (after the deal)

A behaviour is unobservable **ex post**: how carefully an insured party acts, whether a firm really maintains its abatement equipment. Leads to **moral hazard**.

Adverse selection

The classic logic (Akerlof's "market for lemons")

Used cars are "peaches" or "lemons"; only sellers know which. Buyers, unable to tell them apart, will pay only the **average** value.

- At the average price, **peach owners withdraw**; their cars are worth more.
- The pool worsens, the average value falls, the price falls further. . .
- In the limit the **good product is driven out**: the market unravels.

Key insight

Asymmetric information can cause an **entire efficient market to disappear**, even when willing buyers and sellers of high quality both exist.

Lemons in environmental markets

Unobservable environmental quality

Many “green” attributes are exactly the hidden quality of Akerlof’s model: you cannot inspect them at the point of sale.

- **Green claims / greenwashing.** If “sustainable”, “carbon-neutral”, or “recyclable” cannot be verified, credible green products are undercut by cheap imitators, and the green market unravels.
- **Contaminated land.** A seller knows the contamination history; a buyer does not. Honest clean sites struggle to command a premium, weakening incentives to remediate.

Moral hazard

The mechanism (hidden action)

Once a contract shifts risk away from a party, that party has **weaker incentives to take care**, and the change in behaviour is hard to observe or price.

Environmental examples

- **Insurance and precaution.** A fully insured operator may invest less in spill prevention or flood mitigation, raising the chance of the very loss insured against.
- **Clean-up liability.** If a firm expects taxpayers or a bankruptcy shield to cover remediation, it under-invests in safe disposal; abandoned mines and orphaned wells are textbook cases.

Remedies

Deductibles, performance bonds, and **rehabilitation bonds** (required of Australian mines) make the actor bear the consequences of their hidden action.

Information remedies

The idea

If the failure is missing information, the cure is often to **supply credible information**, restoring symmetry rather than restricting the market.

- **Mandatory disclosure.** Pollutant inventories, product-content labelling, contaminated-land registers.
- **Eco-labels and certification.** A trusted third party verifies the hidden quality, breaking the lemons spiral.
- **Standards.** Minimum quality requirements remove the worst “lemons” from the pool.

Government failure

Markets fail, but so can the institutions meant to correct them

- **Rent-seeking:** firms lobby for favourable rules rather than competing.
- **Regulatory capture:** the regulator serves the regulated industry's interests.
- **Unintended consequences:** well-meant rules distort behaviour.
- **Fiscal and informational limits:** governments lack the data and budget to regulate every market.

The Pursuit of Efficiency

When markets fail, what fixes them?

A menu of policy responses

Market failure is a **diagnosis**, not a prescription. Society has two broad routes to restore efficiency:

1. **The courts**: establishing **property rights** and applying **liability rules** so that those who cause harm bear (and weigh) the cost.
2. **Legislation & regulation**: the legislature acts directly:
 - **Command-and-control** standards (mandated technologies / emission limits);
 - **Incentive-based** instruments: charges/taxes, subsidies, **tradable permits**;
 - **Information** policies: disclosure, labelling, registers.

The courts — assigning property rights

Why rights come first

Recall Coase: an externality is really a **missing or ill-defined property right**. Before anyone can bargain, someone must hold the entitlement.

- Courts **establish and enforce** the initial allocation: who has the right to a clean river, quiet air, an unpolluted aquifer.
- Once rights are **clear and enforceable**, and transaction costs are low, the Coase theorem says private bargaining can reach the efficient outcome.

Liability — definition and dual purpose

What is liability?

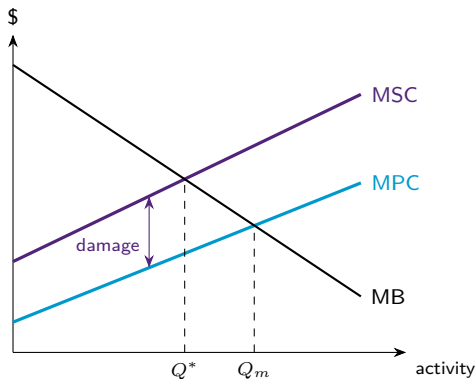
A **liability rule** requires an **injurer** who damages others to **compensate** the victims for the harm caused, typically through court-ordered **damages**.

Two jobs at once

- **Compensation (ex post)**: makes the victim whole, restoring the loss after harm occurs.
- **Deterrence (ex ante)**: the *prospect* of paying damages forces the injurer to **internalise** expected harm *before* acting, and so to take care.

Liability is the legal mechanism by which a private cost rises to meet the **social cost**.

How liability internalises damages



The logic

Without liability the injurer faces only **private cost** (MPC) and over-produces at Q_m .

- A liability rule adds the **expected damage** to the firm's private cost.
- Perceived cost rises toward **MSC**; activity falls toward the efficient Q^* .
- Compensation flows to victims; **deterrence** shrinks the externality.

Two liability regimes

Strict liability

The injurer pays for **all** harm caused, **regardless of fault** or care taken. Common for **abnormally dangerous** activities (toxic chemicals, blasting).

Negligence (fault-based)

The injurer pays **only if** they failed to meet a “**due-care**” **standard**: a legally required level of precaution. Meet the standard and you are **not** liable, even if harm occurs.

Both can, in principle, induce **efficient precaution**, but through different channels.

The incentive to take efficient precaution (1/2)

The decision the injurer faces

Let precaution x cost $c(x)$ and reduce expected damage $D(x)$ (with D falling in x). The efficient level x^* minimises total social cost:

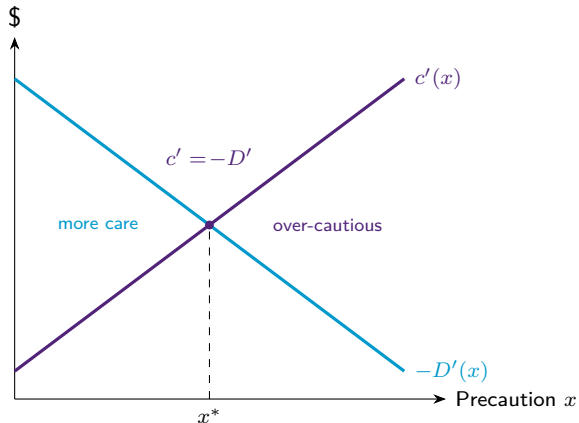
$$\min_x \underbrace{c(x)}_{\text{cost of care}} + \underbrace{D(x)}_{\text{expected harm}} \implies -D'(x^*) = c'(x^*).$$

- **Strict liability:** the injurer bears *both* $c(x)$ and $D(x)$, so they directly minimise the sum $\Rightarrow$ they choose x^* .
- **Negligence:** set the due-care standard $\bar{x} = x^*$. The injurer escapes *all* damages by meeting it, so paying $c(x^*)$ is cheaper than under-caring and facing $D \Rightarrow$ they also choose x^* .

Care is efficient where the **marginal cost of more care** equals the **marginal reduction in expected damage**.

The incentive to take efficient precaution (2/2)

- $c'(x)$ rises: each extra unit of care costs more.
- $-D'(x)$ falls: the first units of care reduce damage a lot; later units less so.
- At x^* : $c'(x^*) = -D'(x^*)$ — one more unit of care costs exactly what it saves in damage.
- $x < x^*$: damage reduction $>$ care cost $\Rightarrow$ more care is worthwhile.
- $x > x^*$: care cost $>$ damage reduction $\Rightarrow$ over-cautious, wasteful.



Both **strict liability** and **negligence** (with $\bar{x} = x^*$) lead the injurer to choose x^* : the same diagram, two routes to the same point.

Where the two regimes diverge

Activity level, not just care

- Under **negligence**, a careful firm pays **nothing** for residual harm, so it has **too little** reason to scale back the risky *activity* itself.
- Under **strict liability**, the firm pays for **all** harm, so it curbs *both* care **and** the activity level, closer to full internalisation.

Information and administration

- Negligence needs courts to **define and verify** the due-care standard, which is demanding and error-prone.
- Strict liability is administratively simpler on *care* but exposes firms to large, uncertain payouts.

Liability rules and the Coase theorem

Two sides of the same coin

A liability rule is just a particular **assignment of entitlements**, backed by a court.

- “Injurer liable” $\equiv$ the victim **holds the right** to be free of harm; the injurer must buy or pay for it.
- “Injurer not liable” $\equiv$ the injurer **holds the right** to act; the victim must pay to reduce harm.
- **Coase**: with **zero transaction costs**, *either* assignment yields the same efficient activity level; only the **distribution** of gains changes.
- With **positive** transaction costs, the choice of liability rule **matters for efficiency**, so courts should place liability on the **least-cost avoider**.

The court process is hard

What a plaintiff must establish

- **Burden of proof:** the victim must prove harm, usually “on the balance of probabilities”.
- **Causation:** link *this* defendant’s emissions to *this* injury, often the hardest step.
- **Probabilistic harm:** pollution raises the *risk* of disease; it rarely leaves a signature on any one victim.
- **Latency:** decades may pass between exposure and illness (asbestos, contaminated water).
- **Many sources:** when dozens of firms emit, **identifying the defendant(s)** and apportioning blame is fraught.

Each hurdle weakens deterrence: harms that cannot be *proved* in court are not fully *internalised*.

Transaction costs and the many-parties problem

Why bargaining and litigation break down

- **Diffuse victims**: air pollution harms thousands a little each, too small individually to sue, a classic **free-rider** and **collective-action** problem.
- **Diffuse sources**: many emitters mean high **coordination** and **attribution** costs.
- **Holdout** and bargaining costs rise sharply with the number of parties.
- **Result**: even where rights are clear, **transaction costs** can swamp the private gains from bargaining.

This is precisely where Coasean bargaining fails, and where **collective**, regulatory solutions look attractive.

The limits of liability

When the deterrent breaks down

- **Bankruptcy / damage caps:** if damages exceed a firm's net worth, the threat is **capped**: harm beyond assets is not deterred.
- **Judgement-proof defendants:** small, under-capitalised, or dissolved firms cannot pay; **insurance** can blunt the very incentive it insures.
- **Scientific uncertainty:** latency, probabilistic harm, and contested causation defeat the burden of proof.
- **High transaction costs:** diffuse, many-party harms are uneconomic to litigate.

When does direct regulation dominate?

Regulation tends to win when harms are . . .

- Common and continuous: routine emissions, not rare accidents;
- High-damage or catastrophic: where *ex post* compensation comes too late;
- Diffuse: many small victims, many sources, high transaction costs;
- Hard to attribute: probabilistic, latent, uncertain causation.

Liability tends to win when harms are . . .

- Rare and discrete, traceable to a solvent, identifiable injurer, and backed by provable causation.

In practice the two are complements: regulation sets ex-ante standards; liability fills the gaps ex post.

Appendix

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